

All Subarray Xors

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, your task is to find all integers that are the xor sum in some subarray.

Input

The first line has an integer n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

First print an integer k : the number of distinct integers that are the xor sum in some subarray.

After this print k integers: the xor sums in increasing order.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq x_i \leq 10^6$

Example

Input	Output
4 5 1 5 9	7 1 4 5 8 9 12 13